

FUNCTIONAL REQUIREMENTS DOCUMENT (FRD)

1. Project Title

Manufacturing Operations Performance Dashboard

2. Dashboard Sections

Overview Section

Executive Operations Overview with KPI summary and trend monitoring.

Drilldown Sections

- Production & Throughput
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3. Data Requirements

Dashboards Section	Data Fields Needed	Source
Overview	Batch Date, Output, Plan, Yield, Scrap, Downtime	Fact Table
Production	Output, Plan, Plant, Downtime	Fact + Dim_Machine
Quality	Scrap Units, QA Score, SKU	Fact + Dim_Product
Downtime	Downtime Minutes, Downtime Reason	Fact + Dim_DowntimeReason
Supplier	Supplier Name, Scrap Rate, QA Fail %	Fact + Dim_Supplier

4. Filters / Slicers

- Batch Date Range
 - Plant City
 - Product Category
 - Shift
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5. Visuals / Charts

Section	Visual Types
Overview	KPI Cards, Line Trend, Heatmap, Bar Chart
Production	Line + Bar Combo, Drilldown Bar Chart, Scatter Plot
Quality	Pareto Chart, Heatmap, Scatter Plot
Downtime	Line Trend, Pareto, Bar Chart
Supplier	Scorecard Bars, Heatmap, Outlier Detection Bar

6. Interactivity

- Cross-filtering between dashboards
 - Drill-down by Plant City → Production Line
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7. Calculations / Measures

Calculation	Formula	Purpose
Plan Attainment %	$\text{SUM}(\text{Actual}) / \text{SUM}(\text{Plan})$	Production efficiency
Yield %	$\text{SUM}(\text{Good Units}) / \text{SUM}(\text{Total Units})$	Quality performance
Scrap Rate %	$\text{SUM}(\text{Scrap Units}) / \text{SUM}(\text{Total Units})$	Loss monitoring
Downtime %	$\text{Downtime} / \text{Total Runtime}$	Operational efficiency
LOD: Plant Yield	$\text{FIXED Plant} : \text{AVG}(\text{Yield \%})$	Plant comparison
Pareto %	$\text{RUNNING_SUM}(\text{Scrap}) / \text{TOTAL}(\text{Scrap})$	Root cause ranking
QA Score	$\text{Yield} - (\text{Scrap Rate} * 2) - \text{Rework Rate}$	Score comparison

8. Notes / Special Instructions

- Star schema implemented with many-to-one relationships.
- Referential integrity assumed during relationship modeling.
- Uniform theme and KPI consistency across dashboards.
- Executive-level readability prioritized.